

$$= 1 - \frac{1}{10} \times \frac{147}{60} = 1 - \frac{147}{600} = \frac{453}{600}.$$

Let the length of the pole be x metres.

$$\text{Then, } \frac{453}{600}x = 12.08 \Leftrightarrow x = \left(\frac{12.08 \times 600}{453} \right) = 16.$$

- 253.** Let the fraction be $\frac{a}{b}$. Then,

$$\begin{aligned} \left(\frac{a}{b} \times \frac{a}{b} \right) \div \frac{b}{a} &= \frac{512}{27} \Leftrightarrow \frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} = \frac{512}{27} \\ \Leftrightarrow \left(\frac{a}{b} \right)^3 &= \left(\frac{8}{3} \right)^3 \Leftrightarrow \frac{a}{b} = \frac{8}{3} = 2\frac{2}{3}. \end{aligned}$$

- 254.** Maximum internal assessment score = $\left(\frac{47}{50} \times 10 \right) = 9.4$.

$$\text{Minimum internal assessment score} = \left(\frac{14}{50} \times 10 \right) = 2.8.$$

$$\therefore \text{Required difference} = (9.4 - 2.8) = 6.6.$$

- 255.** Let savings in N.S.C. and P.P.F. be ₹ x and ₹ $(150000 - x)$ respectively. Then,

$$\frac{1}{3}x = \frac{1}{2}(150000 - x) \Leftrightarrow \frac{x}{3} + \frac{x}{2} = 75000$$

$$\Leftrightarrow \frac{5x}{6} = 75000$$

$$\Leftrightarrow x = \left(\frac{75000 \times 6}{5} \right) = 90000.$$

$$\therefore \text{Savings in Public Provident Fund} = ₹ (150000 - 90000) = ₹ 60000.$$

- 256.** Let there be $(x + 1)$ members. Then,

$$\text{Father's share} = \frac{1}{4}, \text{ share of each other member} = \frac{3}{4x}.$$

$$\therefore 3\left(\frac{3}{4x}\right) = \frac{1}{4} \Leftrightarrow 4x = 36 \Leftrightarrow x = 9.$$

Hence, total number of family members = 10.

- 257.** Let salary = ₹ x . Then, tips = ₹ $\left(\frac{5}{4}x\right)$.

$$\text{Total income} = ₹ \left(x + \frac{5}{4}x \right) = ₹ \left(\frac{9x}{4} \right).$$

$$\therefore \text{Required fraction} = \left(\frac{5x}{4} \times \frac{4}{9x} \right) = \frac{5}{9}.$$

- 258.** Let C's share = ₹ x . Then, B's share = ₹ $\left(\frac{x}{4}\right)$, A's share

$$= ₹ \left(\frac{2}{3} \times \frac{x}{4} \right) = ₹ \frac{x}{6}.$$

$$\therefore \frac{x}{6} + \frac{x}{4} + x = 1360 \Leftrightarrow \frac{17x}{12} = 1360$$

$$\Leftrightarrow x = \left(\frac{1360 \times 12}{17} \right) = ₹ 960.$$

$$\text{Hence, B's share} = ₹ \left(\frac{960}{4} \right) = ₹ 240.$$

- 259.** Let Tanya's share = ₹ x . Then, Veena's share

$$= ₹ \left(\frac{x}{2} \right), \text{ Amita's share} = ₹ \left(\frac{2}{3} \times \frac{x}{2} \right) = ₹ \left(\frac{x}{3} \right).$$

$$\text{Total bill} = ₹ \left(x + \frac{x}{2} + \frac{x}{3} \right) = ₹ \left(\frac{11x}{6} \right).$$

$$\therefore \text{Required fraction} = \left(\frac{\frac{x}{2} \times \frac{6}{11x}}{\frac{11x}{6}} \right) = \frac{3}{11}.$$

- 260.** Let the capacity of the tank be x litres.

$$\text{Then, } \frac{1}{4}x = 135 \Leftrightarrow x = 135 \times 4 = 540.$$

$$\therefore \text{Required fraction} = \left(\frac{180}{540} \right) = \frac{1}{3}.$$

- 261.** Let the capacity of the tank be x litres.

$$\text{Then, } \frac{6}{7}x - \frac{2}{5}x = 16$$

$$\Leftrightarrow 30x - 14x = 16 \times 35$$

$$\Leftrightarrow 16x = 560$$

$$\Leftrightarrow x = 35.$$

- 262.** Let the capacity of the drum be x litres.

$$\text{Then, } \frac{3}{4}x - \frac{7}{12}x = 30$$

$$\Leftrightarrow 9x - 7x = 12 \times 30$$

$$\Leftrightarrow 2x = 360$$

$$\Leftrightarrow x = 180.$$

- 263.** Let x be the number of bottles of oil that the tin can contain.

$$\text{Then, } \frac{5}{8}x - \frac{3}{5}x = 10 - 8 \Leftrightarrow 25x - 24x = 2 \times 40 \Leftrightarrow x = 80.$$

- 264.** R.F. = $\frac{10 \text{ cm}}{1 \text{ m}} = \frac{10 \text{ cm}}{100 \text{ cm}} = \frac{1}{10}.$

- 265.** An area of 49 sq. km is represented by an area of 100 sq. cm on the map.

So, a distance of 7 km is represented by a length of 10 cm on the map.

$$\therefore \text{R.F.} = \frac{10 \text{ cm}}{7 \text{ km}} = \frac{10 \text{ cm}}{7 \times 10^5 \text{ cm}} = \frac{1}{70000}.$$

- 266.** Let the capacity of the bucket be x litres. Then,

$$\text{Capacity of 1 large bottle} = \frac{x}{4}; \text{ Capacity of 1 small bottle} = \frac{x}{7}.$$

$$\text{Fluid left in large bottle} = \left(\frac{x}{4} - \frac{x}{7} \right) = \frac{3x}{28}.$$

$$\therefore \text{Required fraction} = \left(\frac{3x/28}{x/4} \right) = \left(\frac{3x}{28} \times \frac{4}{x} \right) = \frac{3}{7}.$$

- 267.** Let the capacity of 1 bucket = x .

$$\text{Then, capacity of tank} = 25x.$$

$$\text{New capacity of bucket} = \frac{2}{5}x.$$

$$\begin{aligned} \therefore \text{Required number of buckets} &= \frac{25x}{(2x/5)} \\ &= \left(25x \times \frac{5}{2x} \right) = \frac{125}{2} = 62\frac{1}{2}. \end{aligned}$$